

Supporting Information:

Supporting Information: Reading the whole corpus: a reproducible cross-sample Raman workflow for materials characterisation, with a spectral 3D Gaussian splatting representation for hyperspectral maps

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Contents

This document contains the per-map reconstruction figures for every fitted acquisition, the full model and denoiser benchmark tables, and the corpus and knowledge-graph summary tables that the main text quotes selectively. Machine readable versions of every table, the knowledge graph

in GraphML, the fitted fields and the anonymity-gate report are distributed with the released code rather than reproduced here.

Code and companion site. The RAMANSP package and the full pipeline are released at <https://github.com/siddhartha18pahari/RAMANSP>. A companion site at <https://ramansp.vercel.app> carries the same material in browsable form, including an interactive view of the cross-sample knowledge graph, in which the nodes and edges can be dragged, zoomed and inspected one at a time rather than read off a static drawing, together with every figure at full resolution and all three renderings of the manuscript.

Corpus and anonymisation

The corpus holds 108 acquisitions drawn from 3 specimen codes and 16 acquisition configurations. Of these, 57 were decoded directly from the vendor’s binary container; 563 files in an older container variant are not yet handled and are listed in the release. Reading the binaries takes the number of usable hyperspectral maps from 1 to 23. Byte-identical duplicate exports are detected by content hash, and 5 groups were found.

Every specimen identifier, material designation, person name and dated session folder is replaced by an opaque code in a single pass, and the forward map is written to a private file that is excluded from the release. A gate scans everything shareable for surviving identifiers and fails the build on a hit.

Table S1: Composition of the anonymised corpus.

acquisition kind	count
fit table	40
spectrum	36
raw map	23
autocal	7
raw scan	2
total	108

Knowledge graph

Table S2: Knowledge-graph summary. Communities are extracted by greedy modularity maximisation over the content sub-graph only, so provenance edges, which are true by construction, cannot manufacture structure.

quantity	value
nodes	236
edges	636
communities	23
modularity Q	0.65
community purity vs. class	0.86

The graph holds 236 nodes and 636 edges in 23 communities at modularity $Q = 0.65$. The per-acquisition community assignment is released as `tables/kg_communities.csv`, and the graph itself as `graph/knowledge_graph.graphml` with an interactive browser view alongside it.

Spectral 3D Gaussian splatting: all fitted maps

Table S3: Reconstruction across all 6 fitted maps. ρ is primitives per 1000 fitted voxels, and r/σ is the held-out residual divided by that map’s own second-difference noise estimate, so a value at or below one means the field is no further from the data than the data is from itself.

map	N	ρ	compr.	PSNR (dB)	r/σ
map 0059	2904	12	37.7	25.0	0.84
map 0039	2912	13	36.2	22.9	1.02
map 0016	2911	20	23.2	22.6	1.34
map 0028	2912	30	15.5	17.9	1.28
map 0042	2911	32	14.5	23.4	1.18
map 0055	1239	56	8.1	27.2	0.70

Figures S1 onwards give the measured and reconstructed band maps for each fitted acquisition, on a shared colour scale per band with the residual beside them, and the corresponding ellipsoid statistics. The flagship map `acq-0059` appears in the main text and is not repeated here.

Per-map reconstruction figures

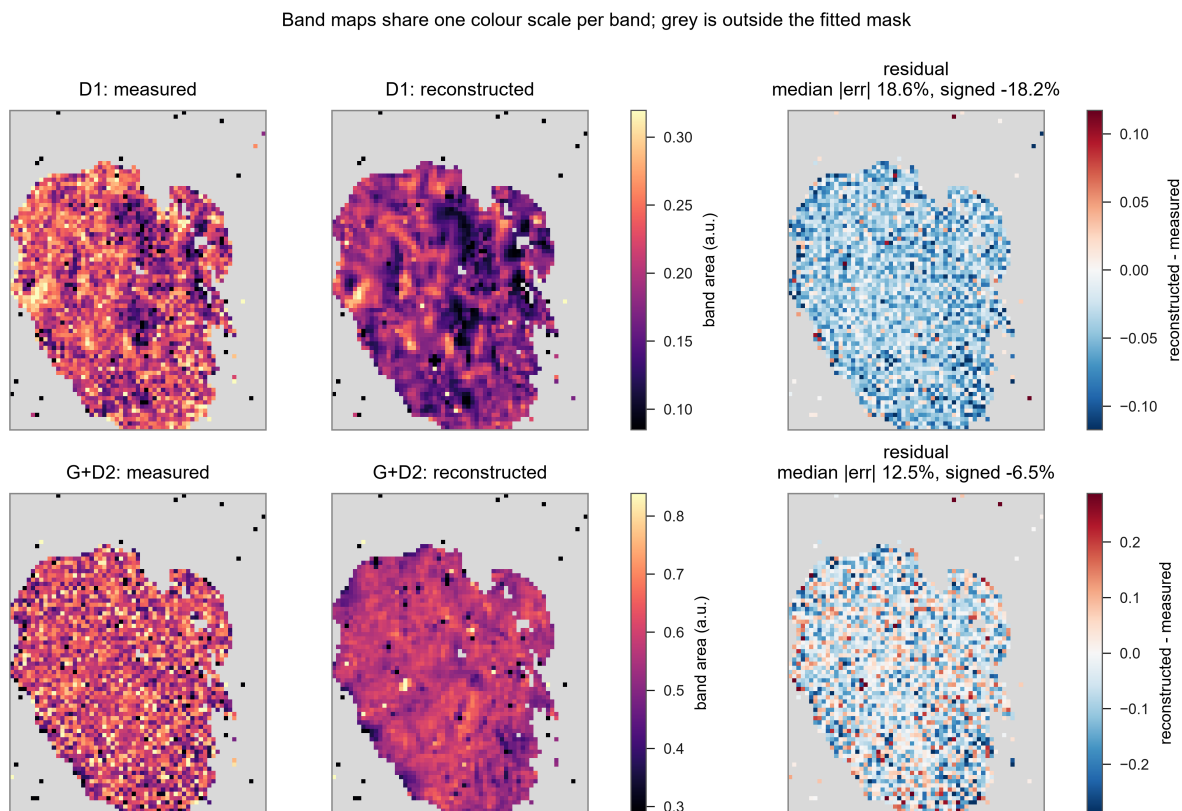
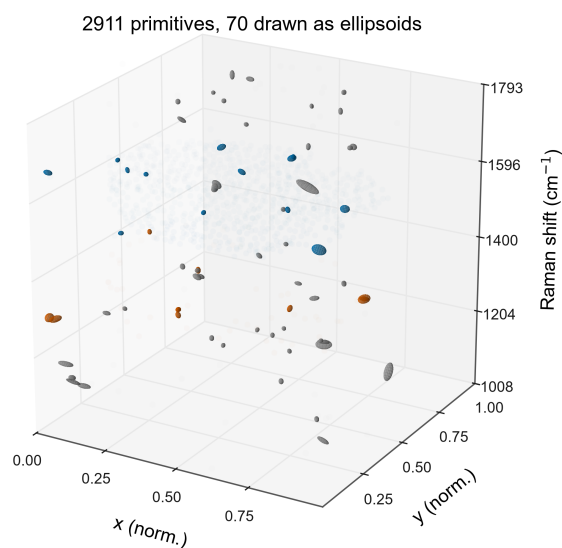
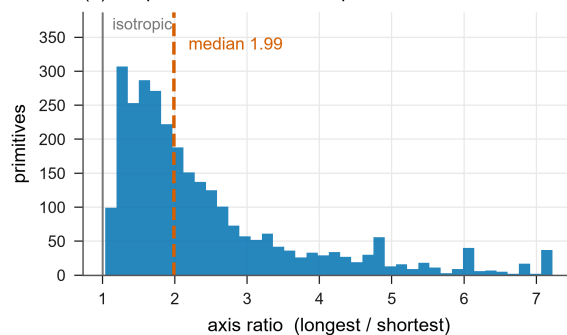


Figure S1: Band maps for acq-0016: measured, reconstructed from the fitted field, and residual, on a shared colour scale per band. Grey is outside the fitted mask.

a



(b) the primitives are anisotropic



(c) amplitude concentrates on the bands

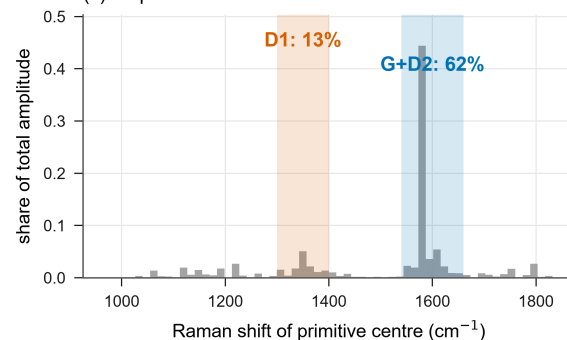


Figure S2: Fitted primitives for acq-0016: the strongest drawn as ellipsoid surfaces, the distribution of axis ratios, and the share of amplitude falling inside each carbon band window.

Band maps share one colour scale per band; grey is outside the fitted mask

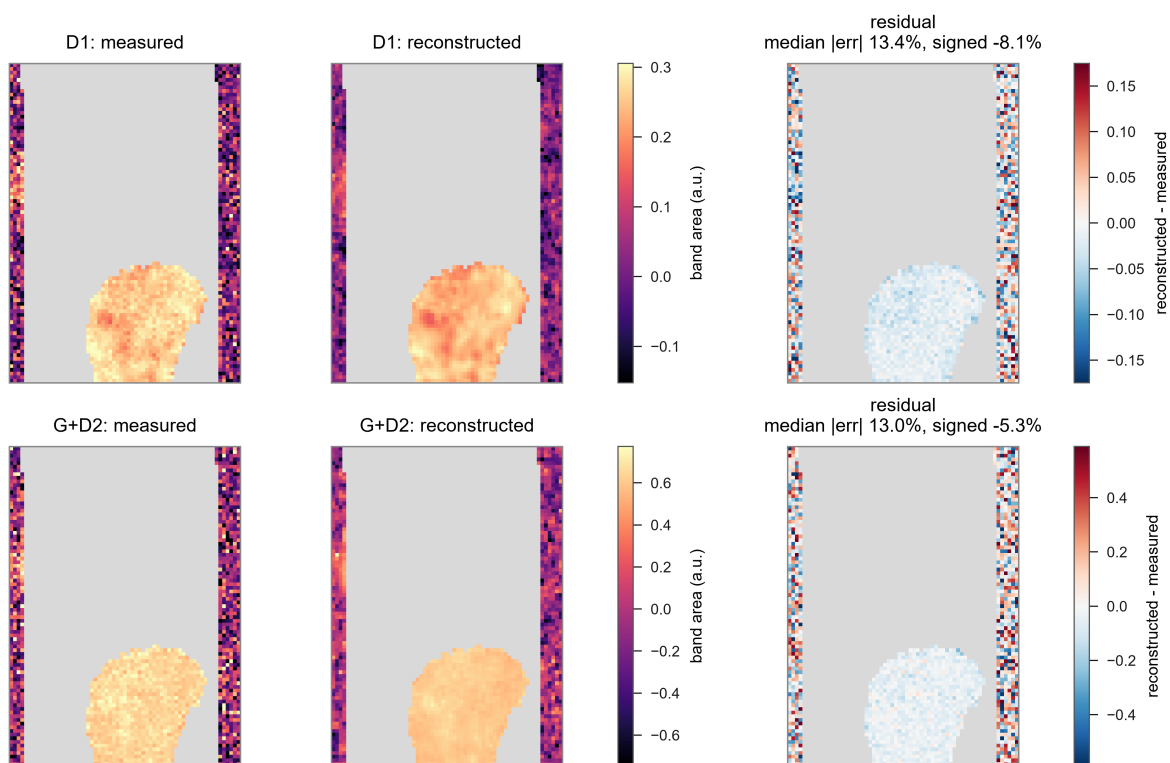


Figure S3: Band maps for acq-0028, as Fig. S1.

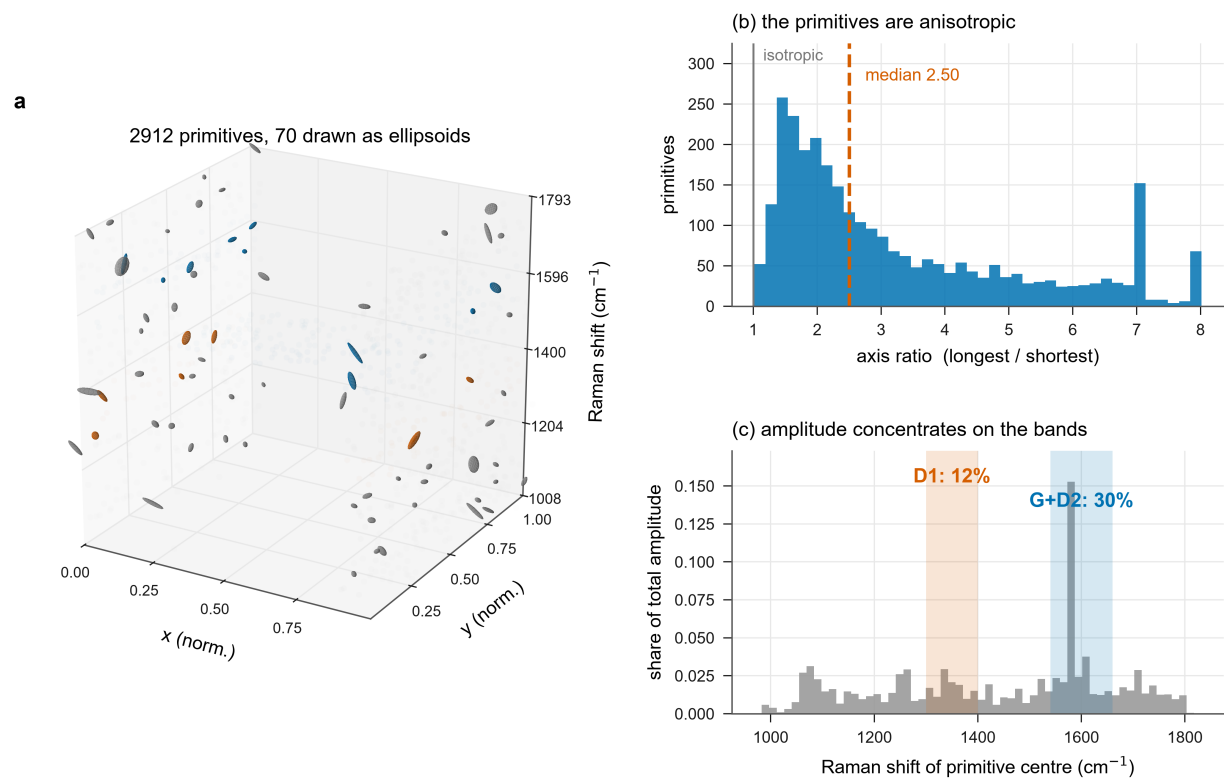


Figure S4: Fitted primitives for acq-0028.

Band maps share one colour scale per band; grey is outside the fitted mask

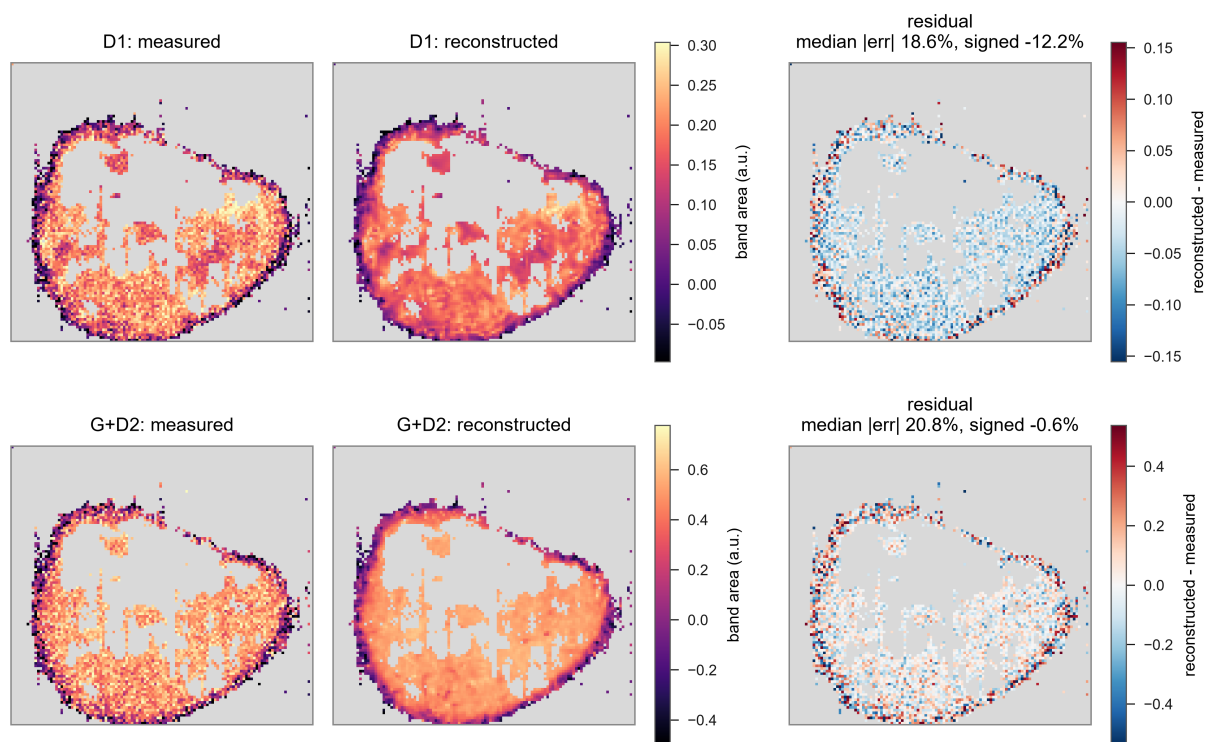


Figure S5: Band maps for acq-0039, as Fig. S1.

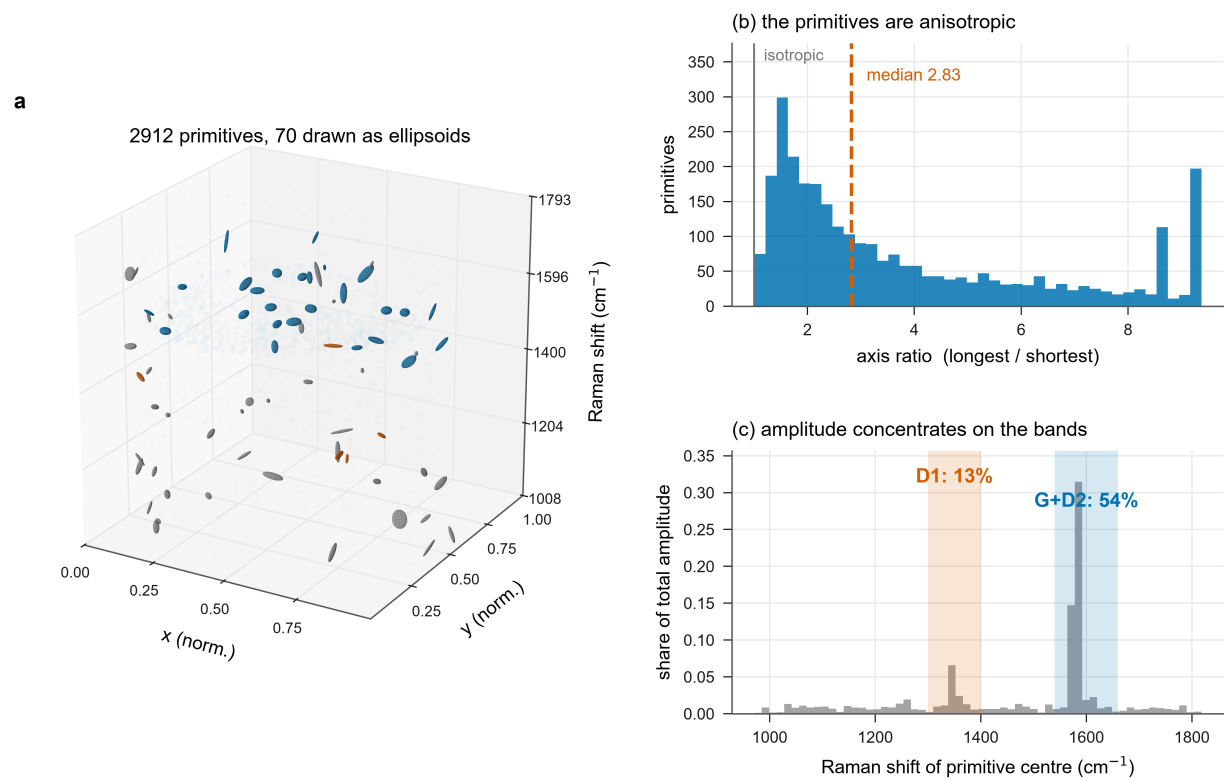


Figure S6: Fitted primitives for acq-0039.

Band maps share one colour scale per band; grey is outside the fitted mask

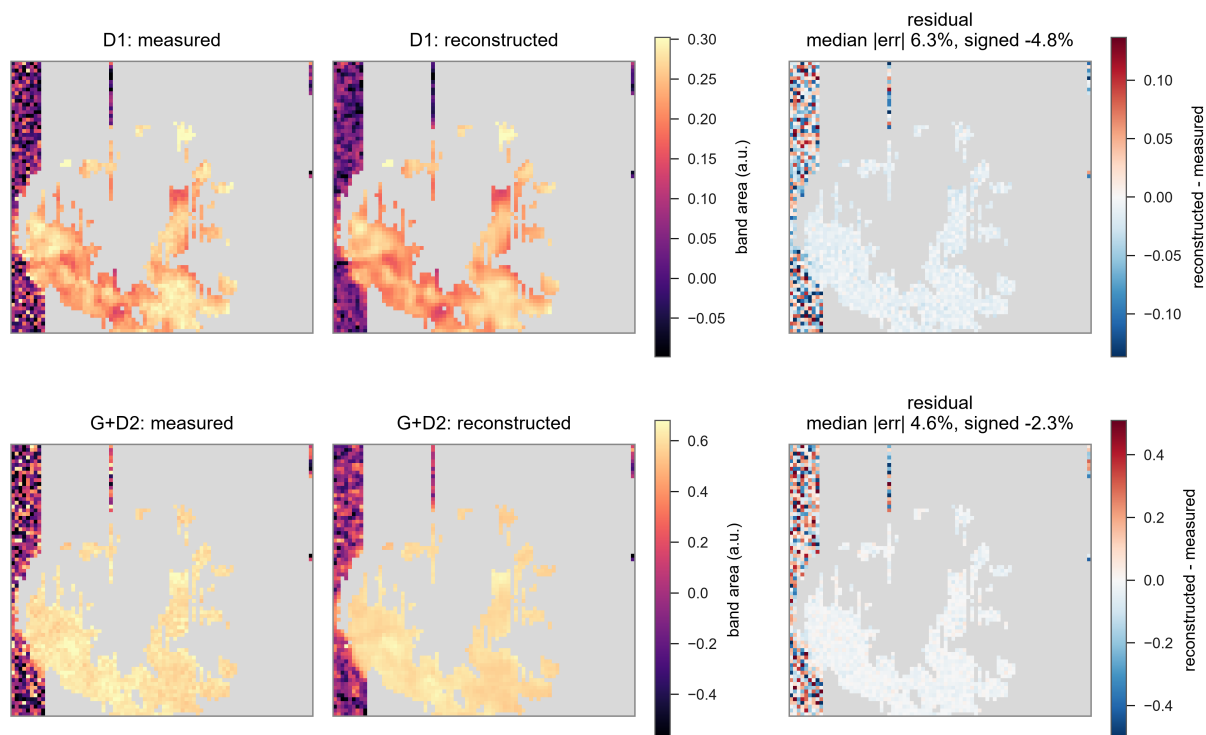


Figure S7: Band maps for acq-0042, as Fig. S1. This is one of the maps on which the original particle mask was inverted; the figure shown is from the corrected mask.

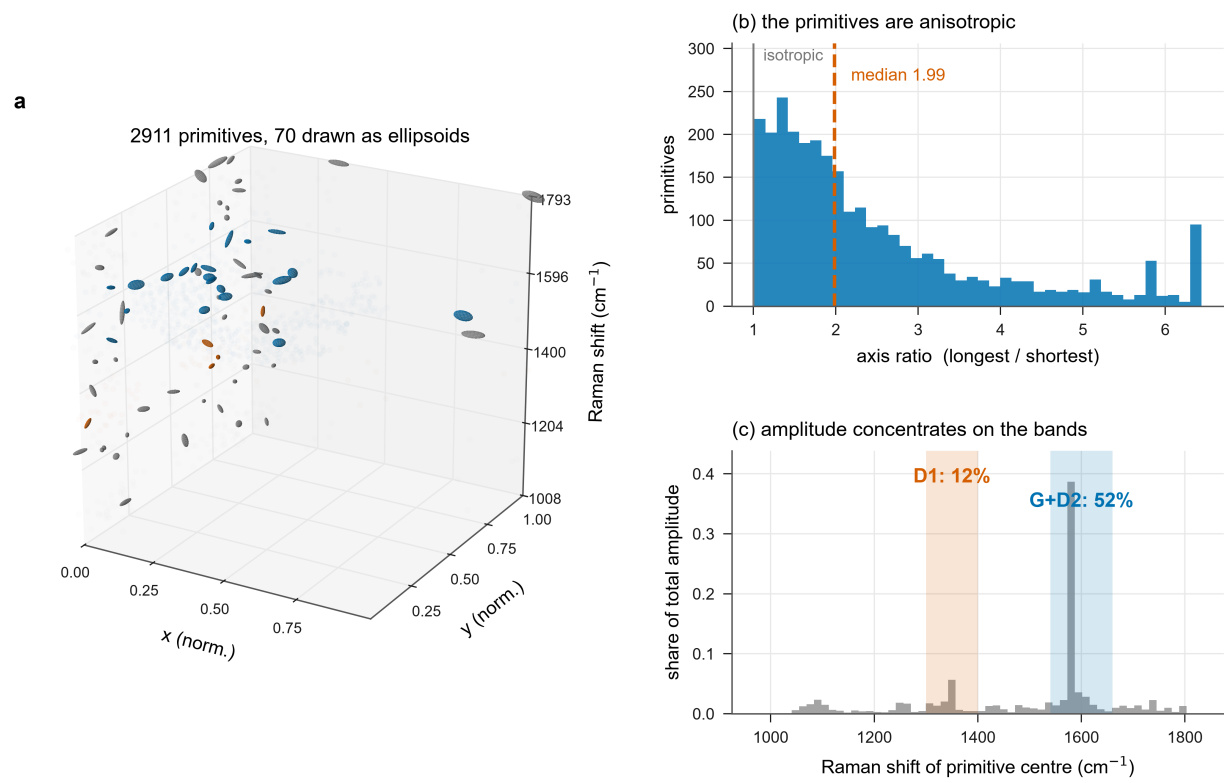


Figure S8: Fitted primitives for acq-0042.

Band maps share one colour scale per band; grey is outside the fitted mask

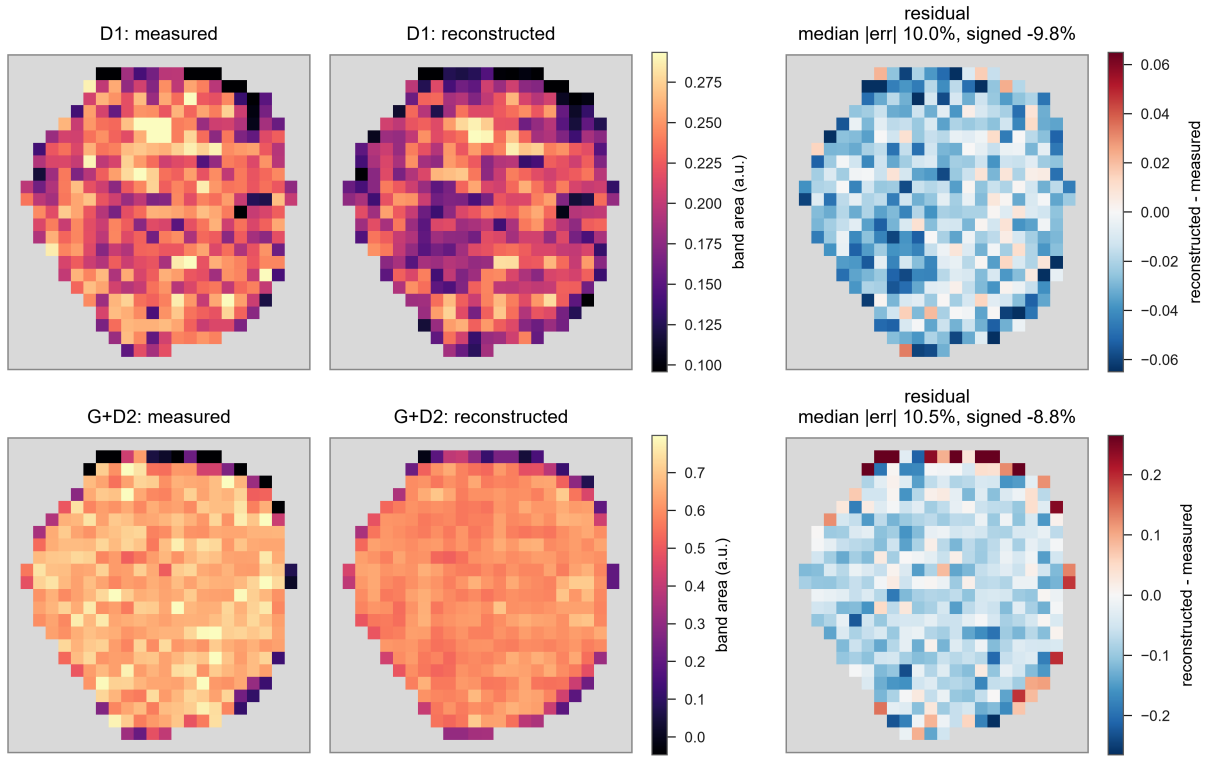


Figure S9: Band maps for acq-0055, as Fig. S1. This is the smallest fitted map and the one whose held-out residual sits furthest below its own noise floor.

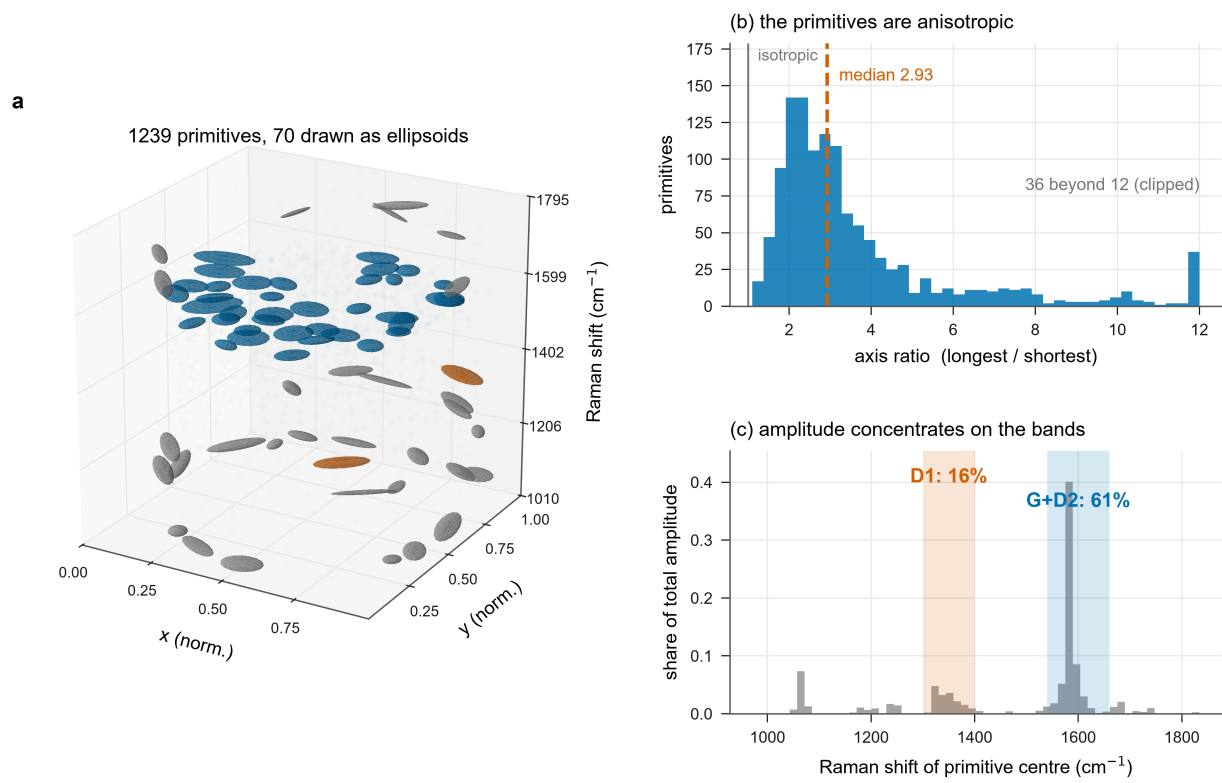


Figure S10: Fitted primitives for acq-0055.

Denoising benchmark, full table

The benchmark contaminates every masked spectrum of the highest contrast-to-noise map, acq-0057, with additive Gaussian noise at 1.5 times that map’s own noise level, and scores 1200 of them against the uncontaminated original. Significance is a two-sided Wilcoxon signed-rank test against SG(2,5), the best Savitzky-Golay filter, with Benjamini-Hochberg correction across all method and metric pairs.

Table S4: Denoising benchmark. Lower is better throughout.

method	MSE	SAD	SID
input (noisy)	1.84e-08	0.0446	0.1340
SG(2,5)	6.71e-08	0.0630	0.1529
SG(3,9)	1.97e-07	0.0943	0.2837
SG(3,15)	6.3e-07	0.2001	0.8262
SG(4,15)	2.9e-07	0.1144	0.4155
SG(3,21)	9.58e-07	0.2862	1.1058
SG(5,25)	7.01e-07	0.2214	0.8947
PCA (rank 217)	1.59e-08	0.0415	0.1127
S3DGS (ours)	7.52e-07	0.1908	0.6960

Classifier benchmark, full table

All 30 classifiers, both tasks, with accuracy, balanced accuracy and weighted F_1 on held-out maps, are released as `tables/ml_models.csv`. Splits are by map, never by spectrum, because spectra from one map share a specimen, a focus, a calibration and a noise realisation and are therefore not independent samples. Models whose cost is quadratic or cubic in the training-set size are fitted on a capped subsample, and the cap is recorded alongside the score in that table so that no comparison is silently unequal.

Preprocessing ablation

The ablation of the main text runs over the 8 smallest maps in the corpus by masked point count, with up to 900 in-mask spectra drawn from each, under 5 named preprocessing protocols. Both limits exist to keep five full preprocessing chains affordable and neither is chosen with reference to the result. The per-map values are released as part of the figure manifest.